

AERE1610H

Project 1: Weight, Balance, & Density Altitude

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Problem 1

Problem statement

Something critical in small aircraft is weight and balance. Instability due to balance or weight issues might result in damage and even death. This problem involves variables and simple arithmetic to check that the weight is meeting the manufacturer's specifications for a 1965 Piper Cherokee PA-28-180.

Theory

$$moment = weight(lbs) * arm(in)$$

$$CG_{aircraft} = \frac{total\ moment}{total\ weight}$$

These are the equations needed to carry out the weight and balance checks.

The first equation is used to get the moment of a force, which can be calculated by multiplying the force by perpendicular distance from pivot to the force's point of action.

The second equation is to obtain the center of gravity of the aircraft. We can calculate this by taking the total moment of the aircraft, which should be able calculated using the previous equation, and dividing it by the total weight of the aircraft.

Solution

In order to solve this problem, I wrote down all my equations, and known variables on a sheet of paper in order to organize everything clearly. Once I found the variables I was missing, I assigned random values to test the weight and balance.

The random values I entered to test the weight and balance were:

- 30 US gallons for amount of fuel
- 100 lbs for pilots weight
- 100 lbs for copilots weight
- 120 lbs for passenger 1 weight
- No passenger 2 (0 lbs)

This is the output that my program generated:

How much fuel is on board (in US Gallons)? 30

fuel =

30

fuel_weight =

180

What is the weight of the pilot (in lbs) 100

weight_pilot =

100

What is the weight of the copilot (in lbs) 100

weight_copilot =

100

What is the weight of passenger one? (0 if no passenger) 120

weight_p1 =

120

What is the weight of passenger two? (0 if no passenger) 0

weight_p2 =

0

The total weight is

empty_moment =

1.2636e+05

```
total_moment =
```

```
1.7473e+05
```

```
new_center_gravity =
```

```
88.6509
```

```
Aircraft is within weight and balance limits>>
```

Discussion

The values I used resulted in a weight less than 2400 pounds and a center of gravity that is between 86.8 and 95.8 inches. This means that the aircraft is within weight and balance limits and therefore able to fly properly and safely.

Weight and balance are very important because they are key factors for an aircraft's structural stability, stress, speed, and takeoff performance. Having an aircraft with proper stability and performance ensures the safety of the people in it, therefore making these factors extremely important in aircraft design.

Problem 2

Problem statement

Another important aspect besides weight and balance is density altitude, which is a representation of what the altitude the aircraft thinks its at depending on temperature, humidity, and pressure.

Density altitude affects life and engine performance, so it is key for pilots as they fly. The lower the number, the better our flight performance.

Theory

$$e = 6.11 \times 10^{\frac{(7.5 \cdot T_d)}{237.7 + T_d}}$$

$$T_d = T_c - \left(\frac{100 - Humidity}{5} \right)$$

$$T_v = \frac{T}{1 - \left(\frac{e}{P_{mb}} \right) * (1 - 0.622)}$$

$$T_{vR} = \left(\frac{9}{5} * (T_v - 273.15) + 32 \right) + 459.69$$

$$D_{alt} = fieldElevation + \left(145366 * \left(1 - \frac{17.326 * P_{inHG}}{T_{vR}} \right)^{0.235} \right)$$

$$P_{inHG} = P_{mb} * 0.02953$$

These are the equations needed to calculate the density altitude.

Equation 1 calculates vapor pressure. The variable Td is dewpoint temperature in Celsius, and the vapor pressure is in millibars .

Equation 2 calculates dewpoint temperature. Tc is the original temperature in Celsius, while Humidity is the current humidity percentage.

Equation 3 calculates virtual temperature. Here, e is vapor pressure, and Pmb is current pressure in millibars. Equation 4 is used to convert virtual temperature from Kelvin to Rankine

Equation 5 calculates density altitude using the previous variables we calculated. fieldElevation is the elevation of the given airport, PinHG is the pressure in inches of Mercury, and TvR is the virtual temperature in Rankine.

The final equation allows us to convert pressure from millibar to inches of Mercury.

Solution

Include

In order to solve this problem I wrote down every single equation we needed before we had to calculate the density altitude. I wrote code to convert every single value that needed to be converted and using those values I was able to plug them in into the density altitude equation.

The values that we were given by openweathermap.org were:

- Temperature: 278 Kelvin
- Pressure: 1021 mb
- Humidity: 72%

This is the output the program generated:

```
temperature =
```

```
278.0600
```

```
pressure =
```

```
1021
```

```
humidity =
```

```
72
```

```
dewpoint =
```

```
-0.6900
```

vapor_pressure =

1.1865

virtualtemp =

278.1822

virtualtempR =

500.7480

pressureinHG =

30.1501

fieldelv =

955.6000

densityaltitude =

-496.1308

Discussion

The density altitude the program calculated was -496.1308 m. As mentioned, this isn't the exact number that the NOAA calculator gives, but it is within 150 feet. The value I got from the NOAA calculator was -1407.7 ft or -427.2 m.

Density altitude is important in aviation because it directly impacts aircraft performance and safety. It tells pilots what the air conditions are compared to a standard.

A higher density altitude means there's less lift and higher takeoff speed is required, while a lower density altitude means there's more lift and better performance.

Appendix

WBcalc.m (Script File Problem 1)

```
%Luis Eduardo Duran AERE1610H Project 1 Problem 1

empty_weight = 1471 %lbs

center_gravity = 85.9 %in

front_seat_arm = 85.5

fuel_tank_arm = 95

rear_seat_arm = 118.1

max_ramp_weight = 2400 %lbs

max_fuel = 50 %US gallons to lbs


%inputs to get missing variables

fuel = input("How much fuel is on board (in US Gallons)? ")

fuel_weight = fuel * 6 %convert to pounds since were using pounds

weight_pilot = input("What is the weight of the pilot? (in lbs) ")

weight_copilot = input('What is the weight of the copilot? (in lbs) ')

weight_p1 = input("What is the weight of passenger one? (0 if no passenger)
")

weight_p2 = input("What is the weight of passenger two? (0 if no passenger)
")


aircraftsafety(fuel, weight_pilot, weight_copilot, weight_p1, weight_p2)
```

aircraftsafety.m (Function File Problem 1)

```
%function script

function[] = aircraftsafety(fuel, weight_pilot, weight_copilot, weight_p1,
weight_p2)

%constants

fuel_weight = fuel * 6 %convert to pounds since were using pounds

empty_weight = 1471 %lbs
```

```

center_gravity = 85.9 %in
front_seat_arm = 85.5
fuel_tank_arm = 95
rear_seat_arm = 118.1
max_ramp_weight = 2400 %lbs
max_fuel = 50 %US gallons to lbs

if fuel > max_fuel
    fprintf("This value is not valid, enter it again ")
    return

end

%compute the total weight and make sure its under the max weight
total_weight = empty_weight + fuel_weight + weight_pilot + weight_copilot +
weight_p1 + weight_p2; % 1 gallon of fuel = 6.073 lbs

fprintf("The total weight is ", total_weight)

if total_weight > max_ramp_weight
    fprintf("Total weight exceeds maximum ramp weight. Please adjust the
weights.\n");
end

%calculate the moments per airplane area and new center of gravity.
front_seat_moment = front_seat_arm * (weight_pilot + weight_copilot);
fuel_tank_moment = fuel_tank_arm * fuel_weight;
rear_seat_moment = rear_seat_arm * (weight_p2 + weight_p1);
empty_moment = empty_weight * center_gravity
total_moment = rear_seat_moment + front_seat_moment + fuel_tank_moment +
empty_moment
new_center_gravity = total_moment / total_weight

```

```

%calculate and make sure we are within the limits for the new center of
%gravity

if new_center_gravity >= 86.8 && new_center_gravity <=95.8
    fprintf("Aircraft is within weight and balance limits")
else
    fprintf("Aircraft is out of weight and balance limits\n");
end
end
end

```

DensityAltCalc.m (Script File Problem 2)

```

%Luis Eduardo Duran AERE1610H Project 1 Problem 2

clear, clc

key = '7cab1fcaf444883263bc48dd983e6018';

options = weboptions('ContentType','json');

url=['https://api.openweathermap.org/data/2.5/weather?
q=','Ames','&APPID=',key];

CurrentData = webread(url, options);

densityaltitude = calcDensityAltitude(key);

```

calcDensityAltitude.m (Function File Problem 2)

```

%function file

function densityaltitude = calcDensityAltitude(city, key)

% given code

key = '7cab1fcaf444883263bc48dd983e6018';

options = weboptions('ContentType','json');

url=['https://api.openweathermap.org/data/2.5/weather?
q=','Ames','&APPID=',key];

CurrentData = webread(url, options);

```

```

%calculations

temperature = CurrentData.main.temp %retrieve temperature

pressure = CurrentData.main.pressure %retrieve pressure

humidity = CurrentData.main.humidity %retrieve humidity

temperatureC = temperature - 273.15; % convert temperature to Celsius

dewpoint = temperatureC - ((100-humidity)/5) %calculate dewpoint temperature

vapor_pressure = 6.11 * 10^((7.5*dewpoint)/(237.7+dewpoint)) %calculate vapor
pressure

virtualtemp = temperature / (1-(vapor_pressure/pressure)*(1-0.622))
%calculate virtual temperature

virtualtempR = ((9/5) * (virtualtemp - 273.15) + 32 ) + 459.69 %convert
virtual temeprature to Rankine

pressureinHG = pressure * 0.02953 %convert pressure in millibars to pressure
in inches of Mercury

fieldelv = 955.6 %given field elevation

densityaltitude = fieldelv + (145336 * (1 - ((17.326 * pressureinHG)/
virtualtempR)^0.235)) %calcualte density altitude using all variables and
calculations

```